Mechanical waves and distant effects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use travel-time reasoning while distinguishing waves from the motion of material.

SUCCESS CRITERIA - I can...

1. Identify the medium carrying the wave.
2. Use speed and distance to estimate travel time.
3. Explain why a simple travel time does not predict damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “medium” and “wave” to explain your answer. Sentence starters are optional.

medium: the material through which a mechanical wave travels

wave: a disturbance that can carry energy and information

signal: a disturbance that carries information

01 Warm-up: bring back an earlier idea

A bundle of parallel rays passes through a lens and becomes **3** times as wide farther beyond it, with rays spreading immediately after the lens. **Name the lens type** and **sketch the rays**.

Word help: lens = a transparent element that changes light-ray directions

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Marzano check - circle: ① ② ③

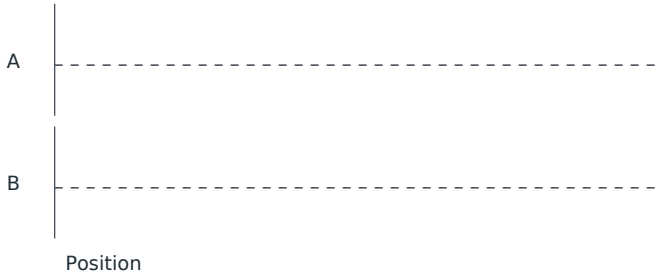
02 Worked example: explain the prediction

Reference: $d = v\,\Delta t$; $\Delta t = d/v$.
Use the supplied wave speed for its medium and wave type. These classroom calculations are not real hazard forecasts.

Worked reasoning: In a simplified description, a wave amplitude falls from **4** to **2** units while intensity is proportional to amplitude squared. Intensity becomes one quarter, not one half.

Your turn: Draw the two waves and two corresponding relative-intensity bars.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

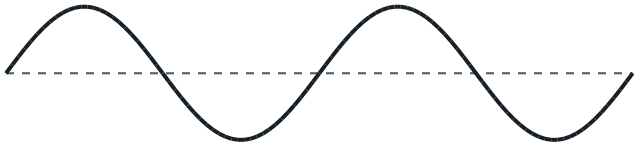


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Draw a disturbance traveling through** water and **mark local water motion**.
2. Why does wave arrival not mean the original water crossed the whole ocean?

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

In a water-wave simulation, a disturbance travels at **20 m/s** between sensors **200 m** apart. **Calculate arrival time**. **Mark the one-way path** and **explain why this** is not the speed of an individual water particle.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why is wave arrival time alone not enough to **predict local damage**? **Name other measurements or observations** you would need.

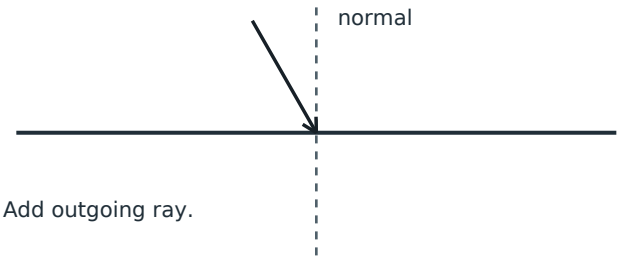
Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A wave enters a medium where its speed is **4** times as large, at a small angle below the critical angle. Does the transmitted ray bend toward or away from the normal? **Label the media**, normal and ray directions.

Sentence starter: In my work, I used … to show … . I labeled it … .

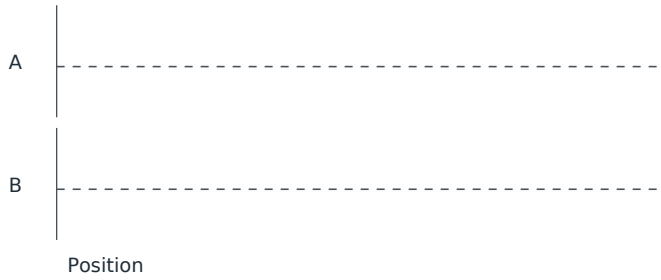


Marzano check - circle: 1 2 3

07 Exit check

At fixed speed, distance doubles. How does travel time change?

Sentence starter: I notice … . This supports … because … .

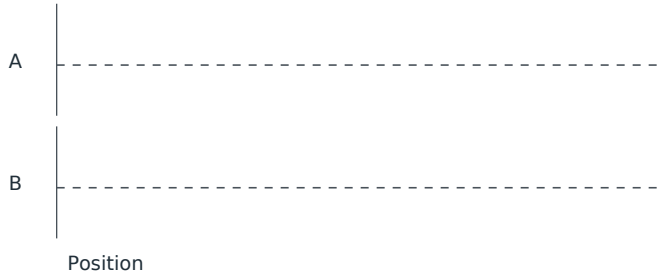


Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two sites have amplitudes **3** and **1** units. **Compare relative intensities** and **name one real-world factor besides** distance that could affect the readings.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3